

## RED WING, WI, USA

WMO: 726564

Lat: 44.589N

Lon: 92.485W

Elev: 780

StdP: 14.29

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 04967

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |       |       |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|-------|-------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |       | 99%   |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB  | DP    | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)   | (g)   | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -13.0      | -8.2 | -20.1                         | 1.9 | -10.4 | -16.8 | 2.3 | -6.5 | 28.6                  | 7.5  | 24.6 | 11.7 | 5.8                   | 300  | 0.531 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 20.9                   | 90.8            | 73.5 | 88.2 | 72.3 | 84.4 | 70.0 | 77.5                | 86.8 | 75.3 | 83.6 | 73.3 | 81.6 | 9.5                  | 200  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 74.7                            | 133.7 | 83.6 | 72.4 | 123.7 | 80.9 | 70.5 | 115.5 | 78.6 | 41.3          | 87.4 | 39.2 | 84.3 | 37.4 | 80.8 | 86.9              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |       |     |
|-------------------|------|------|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|-------|-----|
|                   |      |      | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |       |     |
| 1%                | 2.5% | 5%   | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |       |     |
| (a)               | (b)  | (c)  | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |       |     |
| 22.2              | 19.0 | 16.8 | DB                         | -20.0 | 96.1               | 6.0 | 3.1                                                | -24.3 | 98.4       | -27.8 | 100.2      | -31.2 | 102.0      | -35.6 | 104.3 | (4) |
|                   |      |      | WB                         | -20.4 | 80.4               | 6.0 | 2.6                                                | -24.7 | 82.3       | -28.2 | 83.8       | -31.5 | 85.2       | -35.8 | 87.1  | (5) |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 46.3   | 15.9  | 20.1  | 32.6  | 46.4  | 58.2  | 68.3  | 72.9  | 70.2  | 62.9  | 49.7  | 35.3  | 21.9  |
|                                                                                  | DBStd       | 22.21  | 13.28 | 12.73 | 12.40 | 10.11 | 8.91  | 6.90  | 5.90  | 5.83  | 8.82  | 9.58  | 10.72 | 12.09 |
|                                                                                  | HDD50       | 4111   | 1058  | 838   | 552   | 181   | 24    | 0     | 0     | 0     | 9     | 124   | 453   | 872   |
|                                                                                  | HDD65       | 7519   | 1523  | 1258  | 1004  | 561   | 248   | 44    | 7     | 17    | 143   | 485   | 892   | 1337  |
|                                                                                  | CDD50       | 2773   | 0     | 0     | 14    | 75    | 279   | 549   | 709   | 625   | 395   | 115   | 12    | 0     |
|                                                                                  | CDD65       | 704    | 0     | 0     | 1     | 5     | 38    | 143   | 251   | 177   | 79    | 10    | 0     | 0     |
|                                                                                  | CDH74       | 6784   | 0     | 0     | 7     | 68    | 450   | 1350  | 2551  | 1557  | 695   | 106   | 0     | 0     |
|                                                                                  | CDH80       | 2232   | 0     | 0     | 1     | 16    | 149   | 432   | 948   | 466   | 195   | 25    | 0     | 0     |
| Wind                                                                             | WSAvg       | 6.7    | 7.0   | 7.1   | 7.3   | 8.1   | 7.3   | 6.3   | 5.3   | 5.1   | 6.0   | 7.1   | 7.0   | 6.8   |
| Precipitation                                                                    | PrecAvg     | 34.0   | 1.0   | 0.8   | 2.0   | 3.0   | 3.9   | 5.2   | 4.1   | 5.0   | 4.0   | 2.6   | 2.0   | 1.2   |
|                                                                                  | PrecMax     | 40.7   | 2.3   | 1.7   | 5.1   | 5.9   | 8.6   | 10.0  | 9.7   | 9.3   | 9.3   | 6.6   | 6.7   | 2.8   |
|                                                                                  | PrecMin     | 28.4   | 0.3   | 0.2   | 0.3   | 0.8   | 0.7   | 2.0   | 1.3   | 1.7   | 0.9   | 0.9   | 0.1   | 0.2   |
|                                                                                  | PrecStd     | 3.6    | 0.6   | 0.4   | 1.2   | 1.5   | 1.9   | 2.1   | 1.9   | 2.1   | 2.4   | 1.6   | 1.6   | 0.7   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 45.5  | 53.6  | 72.1  | 81.6  | 90.6  | 92.8  | 95.4  | 92.8  | 90.4  | 83.5  | 68.0  | 53.7  |
|                                                                                  |             | MCWB   | 39.1  | 46.7  | 58.6  | 61.2  | 67.9  | 73.8  | 77.3  | 74.0  | 71.3  | 66.6  | 55.7  | 49.6  |
|                                                                                  | 2%          | DB     | 39.3  | 45.4  | 63.1  | 73.3  | 83.8  | 89.4  | 91.1  | 88.3  | 84.5  | 75.4  | 61.5  | 45.4  |
|                                                                                  |             | MCWB   | 35.5  | 40.4  | 53.4  | 56.0  | 63.9  | 71.7  | 75.3  | 73.4  | 69.6  | 61.9  | 52.3  | 40.9  |
|                                                                                  | 5%          | DB     | 36.4  | 41.0  | 56.4  | 69.6  | 79.2  | 84.3  | 88.4  | 84.2  | 81.5  | 70.4  | 55.4  | 41.0  |
|                                                                                  |             | MCWB   | 33.6  | 36.8  | 48.2  | 53.6  | 62.1  | 68.7  | 73.8  | 71.5  | 67.4  | 58.6  | 47.6  | 37.2  |
|                                                                                  | 10%         | DB     | 34.1  | 37.1  | 50.1  | 63.7  | 73.3  | 81.5  | 84.4  | 81.8  | 78.7  | 64.8  | 51.8  | 37.0  |
|                                                                                  |             | MCWB   | 32.0  | 33.9  | 44.2  | 50.4  | 59.2  | 67.5  | 71.1  | 69.5  | 66.2  | 55.2  | 45.2  | 34.3  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 40.2  | 46.3  | 61.8  | 64.3  | 72.2  | 77.9  | 81.7  | 79.0  | 75.3  | 71.4  | 59.0  | 49.6  |
|                                                                                  |             | MCDB   | 45.0  | 51.3  | 69.2  | 78.0  | 84.7  | 88.0  | 89.4  | 87.3  | 84.5  | 79.3  | 65.4  | 53.9  |
|                                                                                  | 2%          | WB     | 36.5  | 41.3  | 55.1  | 59.3  | 67.9  | 74.9  | 78.7  | 76.2  | 72.4  | 64.5  | 53.6  | 41.2  |
|                                                                                  |             | MCDB   | 39.8  | 45.4  | 61.6  | 70.8  | 78.4  | 83.7  | 88.5  | 84.2  | 81.0  | 72.5  | 59.6  | 44.8  |
|                                                                                  | 5%          | WB     | 33.9  | 37.5  | 49.2  | 55.4  | 65.4  | 72.5  | 76.2  | 74.2  | 70.2  | 60.5  | 49.0  | 37.5  |
|                                                                                  |             | MCDB   | 36.1  | 40.7  | 55.9  | 65.4  | 74.5  | 81.1  | 85.0  | 81.6  | 77.8  | 67.7  | 55.3  | 40.3  |
|                                                                                  | 10%         | WB     | 31.8  | 34.4  | 43.9  | 52.2  | 62.5  | 70.1  | 73.9  | 72.2  | 67.7  | 56.8  | 45.5  | 34.9  |
|                                                                                  |             | MCDB   | 33.9  | 37.4  | 49.1  | 62.1  | 70.7  | 78.8  | 82.4  | 79.7  | 75.8  | 64.9  | 50.1  | 37.4  |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 16.5  | 17.5  | 17.8  | 20.9  | 21.5  | 20.3  | 20.9  | 20.2  | 21.3  | 19.6  | 16.0  | 14.4  |
|                                                                                  |             | MCDBR  | 17.4  | 17.7  | 25.1  | 29.7  | 27.6  | 24.4  | 22.9  | 22.4  | 23.9  | 26.9  | 23.1  | 17.5  |
|                                                                                  |             | MCWBR  | 14.4  | 13.7  | 16.2  | 15.8  | 12.5  | 10.8  | 10.4  | 10.5  | 11.9  | 14.4  | 15.1  | 14.0  |
|                                                                                  | 5% WB       | MCDBR  | 16.2  | 16.4  | 23.3  | 24.4  | 22.7  | 20.3  | 20.2  | 19.5  | 20.1  | 21.4  | 20.7  | 16.0  |
|                                                                                  |             | MCWBR  | 14.0  | 13.2  | 16.0  | 15.1  | 12.1  | 11.1  | 10.7  | 11.1  | 11.6  | 13.7  | 14.6  | 13.3  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.261  | 0.295 | 0.323 | 0.373 | 0.406 | 0.404 | 0.397 | 0.403 | 0.372 | 0.325 | 0.289 | 0.266 |       |
|                                                                                  | taud        | 2.409  | 2.304 | 2.363 | 2.298 | 2.252 | 2.298 | 2.335 | 2.327 | 2.384 | 2.501 | 2.507 | 2.436 |       |
|                                                                                  | Ebn at Noon | 277    | 283   | 289   | 282   | 274   | 275   | 275   | 269   | 269   | 270   | 265   | 263   |       |
|                                                                                  | Edh at Noon | 23     | 31    | 34    | 39    | 42    | 41    | 39    | 38    | 33    | 25    | 21    | 20    |       |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 549    | 890   | 1208  | 1450  | 1683  | 1877  | 1985  | 1663  | 1302  | 833   | 534   | 385   |       |
|                                                                                  | RadStd      | 60     | 98    | 109   | 157   | 133   | 148   | 68    | 109   | 90    | 111   | 51    | 37    |       |

## Historical Trends

|      |                         | DBAvg | Heating |        |       | Cooling |       |     | Degree-Days |       |       |       |
|------|-------------------------|-------|---------|--------|-------|---------|-------|-----|-------------|-------|-------|-------|
|      |                         |       | 99% DB  | 99% DP | 1% DB | 1% WB   | 1% DP |     | HDD50       | HDD65 | CDD50 | CDD65 |
| (46) | Station Only            | (d)   | (e)     | (f)    | (g)   | (h)     | (i)   | (j) | (k)         | (l)   | (m)   |       |
| (47) | Regional (52 neighbors) | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A   | N/A   |       |

Nomenclature: See separate page